Public Key Infrastructure (PKI) and TLS 1.3 Handshake Mechanics

Transport Layer Security (TLS) safeguards confidentiality, data integrity, and mutual authentication across internet communications. Public Key Infrastructure (PKI) establishes trust between parties through hierarchical Certificate Authorities (CAs). A server proves ownership of an identity through an X.509 digital certificate signed by an intermediary CA whose root is trusted in the client's local trust store.

TLS 1.3 substantially improved both speed and security compared to TLS 1.2 by eliminating insecure cryptographic primitives (such as static RSA key exchange and CBC-mode ciphers) and streamlining the initial handshake from two round trips down to a single round trip (1-RTT). The client sends supported cipher suites alongside a Diffie-Hellman key share in the ClientHello. The server selects the cipher, generates its own key share, computes the shared symmetric key, and transmits encrypted handshake extensions.